

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Electron Configuration — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. 2, 8 and 8.
2. s: 2. p: 6.
3. Group 1: one outer electron. Group 7: seven outer electrons.

B

4. N 2,5; Mg 2,8,2; Cl 2,8,7; K 2,8,8,1.
5. O $1s^2 2s^2 2p^4$; Al $1s^2 2s^2 2p^6 3s^2 3p^1$; Ca $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$.
6. They already have a full outer shell, so they do not easily lose, gain or share electrons.

C

7. O^{2-} 2,8 / $1s^2 2s^2 2p^6$; Al^{3+} 2,8 / $1s^2 2s^2 2p^6$; Cl^- 2,8,8 / $1s^2 2s^2 2p^6 3s^2 3p^6$.
8. Calcium is 2,8,8,2. After 2,8,8 the next two electrons occupy the fourth shell, not a 10-electron third shell in this course.
9. Mg is 2,8,2. Losing two electrons leaves 2,8, a full outer shell.

D

10. Mg^{2+} is 2,8 or $1s^2 2s^2 2p^6$. This matches a noble-gas / full outer-shell arrangement, which is stable.
11. Oxygen is 2,6. Gaining two electrons gives 2,8, a full outer shell.
12. Na s-block; F p-block; Fe d-block. Sodium's outer electron is in an s subshell / fluorine is filling 2p / iron is filling 3d.

E

13. P is $[Ne] 3s^2 3p^3$. P^{3-} is $[Ne] 3s^2 3p^6$. The ion has gained three electrons to fill 3p.